

SIMATS ENGINEERING

C04 ASSESSMENT TOOL 1 - DATA INTERPRETATION

Course Name: COMPUTER VISION

Course Code: ITA05

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Questions with Answers – 15 Questions

1. Real-Time System Design Trade-off

Answer: Viola-Jones is the fastest method, but its accuracy is only 78%. YOLO provides 92% accuracy with moderate speed, while Deep Learning gives 95% accuracy but is slow. Under strict latency constraints, **YOLO is the best choice**. It provides a strong balance between speed and accuracy. ViolaJones may be selected only when extremely low latency is more important than accuracy, while Deep Learning is unsuitable when strict real-time response is required.

2. Robust Detection under Environmental Variations

Answer: Deep Learning performs best in all three conditions: 95% in normal conditions, 85% under occlusion, and 80% in low light. YOLO is the second-best, while Viola-Jones drops significantly under difficult conditions. Therefore, **Deep Learning is the most robust overall**. Its main disadvantage is higher computational cost, so hardware acceleration may be required for real-time deployment.

3. Optical Flow Reliability vs Noise Sensitivity

Answer: Method A produces stable motion vectors and is less affected by noise, but it may miss small movements. Method B detects fine motion but produces noisy vectors. For dynamic-scene tracking where reliable trajectories are important, **Method A is generally preferable**. If detecting very small movements is the primary requirement, Method B can be used with filtering or smoothing to reduce noise.

4. Background Modeling Strategy (GMM)

Answer: GMM works well for modeling backgrounds, but it has limitations. Dynamic backgrounds can produce false positives, while static-background situations can cause missed detections because the model may not adapt properly. Therefore, **GMM alone is not sufficient**. It can be improved using adaptive learning rates, shadow removal, morphological filtering, frame differencing, or a more advanced background-subtraction method.

5. Motion Estimation vs Computational Cost

Answer: Method A has a lower error of 5% but requires high computation. Method B has a higher error of 10% but requires less computation. For **real-time applications, Method B is more suitable** because low computation gives faster response. For offline applications, Method A is preferable because accuracy is more important and computation time is less restrictive.

6. Structure from Motion Consistency

Answer: Dataset A gives accurate depth but requires high-quality images. Dataset B works with low-quality images but produces inconsistent depth. For real-world deployment, **Dataset A is preferable when reliable depth accuracy is critical**, provided good cameras and image quality are available. Dataset B is more practical in low-quality environments but requires additional filtering and consistency checks.

7. Stereo Vision vs Motion Estimation

Answer: Stereo Vision provides more accurate depth but requires two cameras. Motion Estimation is less accurate but works with a single camera. Under strict hardware constraints, **Motion Estimation is preferable** because it reduces hardware requirements and cost. If accurate depth is more important and two cameras are available, Stereo Vision should be selected.

8. Model Generalization vs Overfitting

Answer: Model A achieves 98% training accuracy but only 70% testing accuracy, showing strong overfitting. Model B achieves 90% training accuracy and 88% testing accuracy, showing much better generalization. Therefore, **Model B is more reliable for deployment**. Its small training-testing gap indicates that it is less dependent on the training data and is likely to perform better on unseen data.

9. YOLO Performance across Scales

Answer: YOLO performs very well on large objects at 95%, reasonably on medium objects at 85%, but poorly on small objects at 60%. Thus, **YOLO alone is not sufficient for reliable multi-scale detection**. Improvements can include multi-scale feature pyramids, higher input resolution, small-object augmentation, better anchor settings where applicable, and training with more small-object examples.

10. Motion Model Selection for Animation

Answer: Model A is smooth and fast but physically inaccurate, while Model B is realistic but computationally intensive. For **real-time animation**, **Model A is preferable** because responsiveness is important. For cinematic or high-quality offline animation, **Model B is preferable** because realistic motion is more important than computation time.

11. Detection Model Comparison under Resource Constraints

Answer: Model A has 85% accuracy and 40 ms latency, Model B has 92% accuracy and 90 ms latency, and Model C has 90% accuracy and 60 ms latency. For a real-time system with moderate accuracy requirements, **Model A is optimal** because it has the lowest latency and still provides acceptable accuracy. If slightly higher latency is acceptable, Model C is a useful compromise.

12. Robustness vs Speed Trade-off

Answer: Method A has high robustness and 88% accuracy but is slow. Method B is fast with 85% accuracy and moderate robustness. Method C has 90% accuracy but low robustness and moderate speed. For a balanced system, **Method A is the safest choice when robustness is important**. If speed is the main constraint, Method B is preferable. Method C is suitable only when its environmental limitations can be controlled.

13. Optical Flow Stability Analysis

Answer: Method A has high accuracy but high noise sensitivity, while Method B has moderate accuracy and low noise sensitivity. In a noisy environment, **Method B is more suitable for real-time tracking**

because stable output is more important than maximum accuracy. Method A can become unreliable unless a denoising or filtering stage is added.

14. Motion Estimation Accuracy vs Efficiency

Answer: Method A has the lowest error of 4% but high computation. Method B has 9% error and low computation. Method C has 6% error with moderate computation. For embedded systems, **Method C is the best overall choice** because it offers a good balance between accuracy and computational efficiency. Method B is better when resources are extremely limited, while Method A is better when accuracy is the highest priority.

15. GMM Performance in Different Conditions

Answer: GMM achieves 92% accuracy in static environments but drops to 68% in dynamic environments. This shows that GMM is effective for relatively stable backgrounds but is not equally reliable in all environments. Therefore, **GMM alone is not reliable for all conditions**. Improvements include adaptive background modeling, dynamic learning rates, shadow suppression, morphological operations, camera-motion compensation, and combining GMM with object-detection methods.